

AKUNE, Japan

WMO: 478230

Lat: 32.0278N

Lon: 130.2000E

Elev: 138

StdP: 14.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.7	36.6	21.5	16.2	39.0	23.8	18.1	40.5	21.9	47.4	19.7	46.9	9.4	70	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	89.7	78.9	88.0	78.9	86.6	78.5	81.0	86.1	80.1	85.3	79.4	84.6	7.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.6	154.5	84.3	78.8	150.0	83.6	77.9	145.9	83.1	44.8	86.3	43.9	85.6	43.2	84.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	15.9	14.0		DB	31.2	93.9	2.0	1.3	29.8	94.8	28.6	95.6	27.5	96.3	26.0	97.3
				WB	27.8	82.1	1.9	1.0	26.4	82.9	25.3	83.5	24.2	84.1	22.8	84.9

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	46.1	47.9	52.8	60.3	67.4	73.2	80.3	82.0	77.1	68.9	59.4	50.2	(6)
(7)		DBStd	13.13	5.12	5.87	5.44	4.93	3.62	3.68	2.96	2.50	3.89	4.58	5.65	5.68	(7)
(8)		HDD50	350	143	103	32	1	0	0	0	0	0	0	2	69	(8)
(9)		HDD65	2272	586	480	380	153	16	0	0	0	0	15	183	459	(9)
(10)		CDD50	5418	22	43	118	311	539	696	941	992	813	585	283	75	(10)
(11)		CDD65	1865	0	0	1	13	90	246	476	527	363	135	14	0	(11)
(12)		CDH74	14771	0	0	1	16	188	1082	4618	5685	2704	465	12	0	(12)
(13)		CDH80	4088	0	0	0	0	11	129	1289	1971	642	46	0	0	(13)
(14)	Wind	WSAvg	7.1	8.4	8.4	8.2	7.1	6.0	5.8	6.0	5.7	6.4	7.1	7.4	8.3	(14)
(15)	Precipitation	PrecAvg	84.2	3.1	3.9	5.3	6.3	7.3	17.8	12.6	8.4	7.9	3.8	4.1	3.7	(15)
(16)		PrecMax	135.2	6.2	8.0	13.0	15.6	14.5	40.1	36.5	26.4	17.7	11.2	8.5	6.7	(16)
(17)		PrecMin	52.8	0.7	1.6	2.5	3.2	2.3	3.5	2.0	1.1	1.2	0.4	1.2	0.3	(17)
(18)		PrecStd	18.3	1.2	1.8	2.1	2.9	3.6	7.8	8.1	5.3	5.2	2.8	2.0	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.6	67.6	70.8	76.2	81.5	86.3	91.3	93.2	89.7	84.0	75.6	67.7	(19)
(20)			MCWB	60.0	62.2	61.3	65.6	69.7	77.8	78.6	78.5	77.2	74.7	67.5	61.2	(20)
(21)		2%	DB	60.5	64.2	67.3	73.0	78.1	83.5	88.6	90.7	87.1	81.0	72.9	64.8	(21)
(22)			MCWB	55.7	59.8	60.5	64.5	68.7	76.5	79.2	78.7	77.5	72.4	65.5	59.2	(22)
(23)		5%	DB	57.6	61.3	64.8	70.9	75.9	81.0	87.1	88.8	85.4	78.6	70.6	62.0	(23)
(24)			MCWB	52.1	56.8	59.0	63.8	67.8	75.0	79.1	79.1	76.6	70.0	64.0	55.9	(24)
(25)		10%	DB	54.6	58.0	62.4	68.7	74.2	79.2	85.8	87.4	83.7	76.5	68.2	59.1	(25)
(26)			MCWB	48.9	52.5	56.4	62.0	67.0	74.2	78.7	78.9	75.5	68.6	62.3	53.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	64.3	65.1	70.3	74.3	79.5	82.1	82.5	80.3	77.2	70.7	63.9	(27)
(28)			MCD B	63.7	66.1	67.3	72.5	77.5	84.0	86.6	87.4	85.4	81.5	73.3	65.8	(28)
(29)		2%	WB	56.9	60.6	62.3	67.6	72.0	78.1	80.9	81.3	79.1	74.5	67.6	60.0	(29)
(30)			MCD B	59.8	63.5	65.5	70.6	75.2	81.8	85.5	86.7	84.5	79.2	70.6	63.7	(30)
(31)		5%	WB	53.0	57.4	60.0	65.3	70.2	77.0	80.0	80.5	78.1	72.4	65.5	56.8	(31)
(32)			MCD B	56.7	61.0	64.1	69.1	73.5	80.0	84.7	86.1	83.6	77.0	69.2	60.9	(32)
(33)		10%	WB	49.6	53.4	57.4	63.3	68.4	75.7	79.3	79.8	76.9	70.1	63.1	54.3	(33)
(34)			MCD B	53.8	57.3	61.8	67.4	72.5	78.4	84.0	85.4	82.4	75.0	67.5	58.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.9	12.1	12.9	13.0	12.5	9.7	9.5	10.9	11.5	12.2	11.9	11.3	(35)
(36)			MCD BR	15.2	15.3	15.5	15.2	14.5	11.3	11.4	12.8	12.5	13.8	13.3	13.8	(36)
(37)			MCWBR	12.5	13.3	11.6	10.2	8.9	6.4	5.7	5.9	6.2	7.6	8.9	10.7	(37)
(38)		5% WB	MCD BR	14.5	13.8	13.6	11.9	10.9	8.2	9.8	11.0	11.1	10.9	10.9	12.8	(38)
(39)			MCWBR	13.2	13.6	12.0	10.0	8.5	5.5	5.4	5.6	6.1	7.3	8.9	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.433	0.468	0.524	0.543	0.552	0.554	0.537	0.521	0.502	0.463	0.430	0.423	(40)	
(41)		taud	2.079	2.017	1.882	1.891	1.936	2.020	2.106	2.149	2.155	2.174	2.183	2.112	(41)	
(42)		Ebn at Noon	240	245	241	243	242	240	244	246	245	245	241	236	(42)	
(43)		Edh at Noon	42	49	61	63	61	56	51	48	46	42	38	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	717	941	1278	1521	1598	1323	1672	1712	1413	1176	854	676	(44)	
(45)		RadStd	73	90	85	141	178	215	205	133	148	124	83	62	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page